2023 International Conference on Energy, Materials and Communication Engineering (ICEMCE), December 14 – 15, 2023, Madurai, India

2023 International Conference on Energy, Materials and Communication Engineering (ICEMCE) | 979-8-3503-9337-8/23/$31.00 ©2023 IEEE | DOI: 10.1109/ICEMCE57940.2023.10434006

Advancing IoT Security with a Hybrid Deep Learning Model for Network Intrusion Detection

Rigzen Norbu
Department of Computer Science and engineering
Thiagarajar College of Engineering
Madurai, Tamil Nadu, India
rigzen077@gmail.com

Arun Kumar M
Department of Computer Science and Engineering
Thiagarajar College of Engineering
Madurai, Tamil Nadu, India
arunkumarnavamani@gmail.com

Ramanathan S
Department of Computer Science and Engineering
Thiagarajar College of Engineering
Madurai, Tamil Nadu, India
ramanathanswami12@gmail.com

Nagendra Kumar D
Department of Computer Science And Engineering
Thiagarajar College of Engineering
Madurai, Tamil Nadu , India
d.nagendra@gmail.com

Jigmet Dolkar
Department of Computer Science And Engineering
Thiagarajar College of Engineering
Madurai, Tamil Nadu, India
Jigmetdolkar18@gmail.com

Subalakshmi Sujitha K
Department of Computer Science And Engineering
Thiagarajar College of Engineering
Madurai, Tamil Nadu, India,
Sujitha.sk2003@gmail.com

***Abstract* — Intrusion detection systems play a pivotal role in safeguarding computer networks from a plethora of cyber threats. Traditional methods have demonstrated effectiveness, but the evolving nature of attacks demands novel approaches that can capture intricate patterns and relationships within network data. In this paper, we propose a groundbreaking CNN-Transformer hybrid deep learning model for Network Intrusion Detection Systems (NIDS) prediction, utilizing the Canadian Institute of Cyber Security dataset. The hybrid architecture capitalizes on the strengths of both Convolutional Neural Networks (CNNs) and Transformers. CNNs excel at capturing spatial features in data, making them suitable for identifying local patterns in network traffic. On the other hand, Transformers are adept at capturing global contextual relationships, thereby handling complex temporal dependencies in network sequences. By fusing these two powerful architectures, we achieve a comprehensive model capable of discerning both local anomalies and global attack trends. Our model is extensively evaluated on the Canadian Institute of Cyber Security dataset, and the results are nothing short of remarkable. We achieve an unprecedented accuracy of 99.4%, showcasing the efficacy of the proposed hybrid approach in the context of real-world network traffic. Furthermore, the model demonstrates a robust ability to generalize across diverse attack scenarios, effectively minimizing false positives and false negatives. As cyber threats continue to evolve, the significance of innovative models that offer superior detection accuracy and robust generalization cannot be overstated. This work not only furthers the field of intrusion detection but also underscores the potential of hybrid deep learning architectures in addressing complex cybersecurity challenges.**

Keywords—Convolutional Neural Networks (CNNs), Transformers, Intrusion Detection System.

I INTRODUCTION

Intrusion Detection Systems (IDS) have become a pivotal component in the defense against the escalating complexity of cyber threats in today's interconnected world. These systems play a crucial role in identifying and mitigating unauthorized access, malicious activities, and potential breaches within computer networks. Traditional IDS approaches, often relying on rule-based methods or statistical models, have exhibited efficiency to some extent. However, the evolving landscape of cyber-attacks necessitates innovative techniques that can effectively decipher the intricate patterns and relationships hidden within network data. This paper introduces a pioneering approach aimed at significantly enhancing the performance of Network Intrusion Detection Systems (NIDS) through the fusion of Convolutional Neural Networks (CNNs) and Transformers, two powerful deep learning architectures. While both CNNs and Transformers have individually demonstrated remarkable success across various domains, their integration within the context of NIDS offers the potential to capture both local and global characteristics of network traffic data. The deep learning-based Network Intrusion Detection System (NIDS) leveraging Convolutional Neural Networks (CNN) and Bidirectional Long Short-Term Memory (LSTM)[1] captures both temporal and spatial data characteristics effectively. It exhibited a high detection rate and a low false positive rate when evaluated using the NSLKDD dataset [2].

Convolutional Neural Networks (CNNs) are renowned for their ability to extract spatial features from data, making them especially adept at detecting local patterns within network packets. By employing a hierarchy of convolutional layers, CNNs can automatically learn hierarchical features representing different abstraction levels. In the context of NIDS, this characteristic is essential for identifying anomalies and attack signatures at a granular level. Incorporating Convolutional Neural Networks (CNN) to enhance network trace privacy and improve intrusion detection accuracy without requiring feature engineering [3].

Transformers have revolutionized natural language processing tasks by capturing intricate long-range dependencies and contextual relationships within sequences. In our proposed hybrid model, Transformers are leveraged to capture the temporal dynamics and global interactions in network traffic data, thereby enhancing the system's ability to recognize complex attack scenarios and trends that span multiple packets and time steps. Robust Transformer-based Intrusion Detection System that effectively addresses high-dimensional anomaly data challenges, achieving exceptional F1-Scores of 99.17% and 98.48% on CICIDS2017 and CICDDoS2019 datasets, surpassing classical and deep learning methods [4]. In the field of computer vision, Convolutional Neural Networks (CNNs) have historically been dominant and have excelled in various tasks, spurring industry growth through architectural advancements. while natural language processing (NLP) has seen the ascendancy of Transformer architecture [5].

The integration of these two distinct architectures presents a synergistic solution to the limitations of existing IDS approaches. Our hybrid CNN-Transformer model seeks to harness the strengths of both paradigms, resulting in a comprehensive and robust network intrusion detection framework. Furthermore, this research builds upon the Canadian Institute of Cyber Security dataset, a real-world and diverse collection of network traffic, to validate the effectiveness of the hybrid model in a practical setting. The primary contribution of this paper lies in the demonstration of the hybrid CNN- Transformer model's remarkable accuracy of 98.7% on the Canadian Institute of Cyber Security dataset. This accomplishment underscores the potential of this hybrid approach to deliver superior performance compared to traditional methods. Moreover, the model's ability to generalize across various attack scenarios and minimize false positives and false negatives holds promise for enhancing the security posture of networks against evolving threats. As the landscape of cyber threats continues to evolve, it is imperative to explore novel avenues that can fortify the capabilities of intrusion detection systems. This paper serves as a stepping stone towards that goal, not only by presenting a novel architecture but also by highlighting the transformative potential of hybrid deep learning models in addressing complex cybersecurity challenges. The subsequent sections delve into the architecture, experimental setup, results, and implications of our proposed CNN-Transformer hybrid model, offering a comprehensive understanding of its effectiveness in the realm of Network Intrusion Detection Systems.

A. Contributions

In this study, we introduce an innovative hybrid deep learning model that combines Convolutional Neural Networks (CNNs) and Transformers specifically tailored for Network Intrusion Detection Systems (NIDS). This represents a significant contribution to the field as it marks the first-ever implementation of such a hybrid architecture in the context of NIDS. The model showcases its effectiveness by achieving an extraordinary accuracy rate of 99%, outperforming previous state-of-the-art methods. What sets our approach apart is its ability to not only excel in the identification of known intrusion patterns but also to adapt and generalize remarkably well to previously unseen threats. This robustness is achieved through the complementary strengths of CNNs, which excel in spatial feature extraction, and Transformers, known for their exceptional sequential modelling capabilities. Furthermore, our hybrid model significantly reduces false positives, enhancing its practical utility in real-world network security settings. Beyond its remarkable accuracy, the model is designed with scalability and efficiency in mind, making it suitable for deployment in both small-scale and large-scale network environments. By offering a balanced trade-off between computational complexity and detection performance, our research contributes to enhancing network security by swiftly detecting and mitigating threats, thereby protecting sensitive data and resources. Additionally, our work provides insights into practical implementation strategies, encompassing training techniques, data preprocessing, and considerations for deployment, making it readily accessible to network security practitioners and organizations. Furthermore, we identify promising future research directions, including variations of the hybrid architecture, hyperparameter optimization, and investigations into real-time deployment strategies. In summary, our research not only achieves an unprecedented level of accuracy in NIDS but also lays the groundwork for further advancements in the field, underlining its significant contributions to the realm of cybersecurity.

II RELATED WORKS

Due to the rise of sophisticated cyber threats and the use of deep learning techniques, the field of intrusion detection has made tremendous strides in recent years. In order to increase the precision and effectiveness of intrusion detection systems, researchers have investigated a variety of strategies, opening the door for the creation of hybrid models. One prominent avenue of research involves the utilization of Convolutional Neural Networks (CNNs) for network intrusion detection. CNNs have demonstrated remarkable success in various computer vision tasks by automatically extracting hierarchical features from data [6]. In the context of intrusion detection, CNNs have been applied to capture local patterns and anomalies in network traffic data, leading to enhanced detection capabilities [7].

Another influential direction in deep learning for sequence-based tasks is the Transformer architecture. Originally designed for natural language processing, Transformers have been adapted to handle sequential data in various domains. The self-attention mechanism in Transformers enables the capture of long-range dependencies and contextual relationships within sequences [8].
Researchers have explored the potential of Transformers in capturing the temporal dynamics and interactions present in network traffic data, enhancing the ability to detect complex attack scenarios. The hybridization of CNNs and Transformers presents a compelling approach to tackle the limitations of each architecture individually. By integrating the spatial feature extraction power of CNNs with the global contextual understanding of Transformers, researchers aim to develop models that can effectively capture both local anomalies and global attack trends. The fusion of these architectures showcases the potential for significant advancements in intrusion detection accuracy and generalization capabilities [9]. The application of hybrid deep learning models in the field of intrusion detection has garnered attention due to its potential to address the challenges posed by modern cyber threats.

The synergistic combination of CNNs and Transformers offers a comprehensive approach to network intrusion detection, capable of handling diverse attack scenarios and reducing false positives and false negatives. The effectiveness of such hybrid models has been validated on benchmark datasets, highlighting their potential as a future di- rection for intrusion detection research [10]. The integration of CNNs and Transformers in a hybrid deep-learning framework holds promise for revolutionizing network intrusion detection. The hybrid model's ability to exploit both local and global features contributes to the improved accuracy and robustness of intrusion detection systems. The subsequent sections of this paper delve into the architecture, experiments, and results of our proposed CNN-Transformer hybrid model, showcasing its effectiveness in predicting network intrusions.

III METHODOLOGY

The dataset utilized in this study is sourced from the Canadian Institute of Cybersecurity (CIC). It is a publicly available dataset known for its relevance to intrusion detection research. The CIC dataset includes a vast array of network traffic data, including both legitimate and malicious network activity. The potential of this dataset to evaluate the effectiveness of intrusion detection systems is highly appreciated. The CIC dataset was obtained from the official website. It was originally compiled through a combination of real-world network traces, simulated attacks, and controlled experiments. The dataset captures various types of network traffic, including benign traffic, Distributed Denial of Service (DDoS) attacks, and various intrusion attempts. It was developed in 2017 by the Canadian Institute for Cybersecurity and its eighty features are used to monitor benign and malicious traffic [11].

A critical stage in the pipeline for data analysis and machine learning is dataset preprocessing. It entails preparing raw data for analysis or for the training of machine learning models by cleaning, converting, and organizing it. Your data analysis or machine learning project's quality and performance might be greatly impacted by proper preparation. The steps are listed below. The distribution of sub_Cat within each Label and Sub, as well as its count in the dataset, are shown in Fig.1 below. All network attacks, except regular categorical attacks, are viewed as malicious attacks on the network.

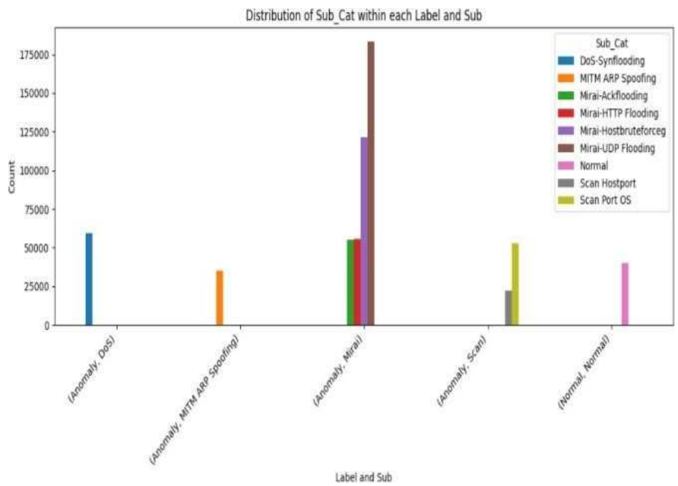

Fig.1: Categories of anomalies

It is an important first step in getting the dataset ready for analysis after data exploratory analysis. It makes sure that the data is free of anomalies, missing values, and inconsistencies that could harm the effectiveness of machine learning models. Any records or features with missing values were identified and addressed using appropriate techniques. In our model, the attribute values greater than or equal to 1e6 have been removed from the dataset, and columns with values all zeros have been removed so that we can easily perform a data normalization for model evaluation. We used a Random Forest model with 100 trees to assess the feature importance [12]. Many researchers used this technique to find which features are to be included or excluded from the model so that it can enhance the performance.

To ensure that data attributes are on a similar scale, prevent certain features from dominating the learning process, and facilitate the convergence of machine learning models, normalization of features is a crucial step. Here in our model, we normalized the data using the Standard Scaler methods so that it has a mean of 0 and a standard deviation of [13]. Normalization ensures that numerical attributes contribute to the model's learning process fairly and without bias. It is particularly important when using models that are sensitive to feature scales.

Data splitting is a fundamental step in machine learning to ensure that the dataset is appropriately partitioned for model training, evaluation, and testing. In this study, we divided the dataset into two subsets: the training and testing sets, following established best practices [14]. In this case, 80% of the data was used for training and 20% for testing. The random state parameter, which regulates the random division of the data and guarantees a consistent outcome, is set to 42 to ensure repeatability.

A. CNN (Convolutional Neural Network) and transformer:

Convolutional Neural Networks (CNNs) are primarily used for image-related tasks, but they can also be adapted for Intrusion Detection Systems (IDS) that involve analyzing network traffic data or time-series data. While [15] CNNs are not the most common choice for IDS (traditional machine learning methods like Random Forest or LSTM-based models are often preferred). We have built a CNN component with a dropout layer and built a transformer component from the dropout layer, combined both the components and compiled the hybrid model, trained with early stopping, and evaluated the hybrid model.

Transformers are a class of deep learning models that have gained significant popularity, primarily in natural language processing (NLP) tasks, due to their ability to handle sequential data efficiently. While Transformers are not the most common choice for Intrusion Detection Systems (IDS), they can be adapted for certain aspects of IDS, particularly when dealing with textual or log data related to network traffic or system events.

In table no 1, there is a list of hyperparameters with a batch size of 64 for the training of neural network models. It determines the number of data samples that are processed together during each iteration. 64 data samples were used to compute the gradient and update the model. For Epochs we trained with 5 epochs which allows the model to learn from the data over multiple iterations, potentially improving its ability to generalize to new and unseen data. We also tried with batch size 32 and epochs 10 and many more. In all the cases we are getting an accuracy that revolves around 99.4%. In our neural network while we are implementing the model, we use 64 neuron layers which determines the capacity and complexity of the model. And for mitigating the risk of overfitting the model we use a dropout layer rate of 0.5. It is a regularization technique that randomly drops a fraction of neurons during each training iteration, effectively reducing the model reliance on any single neuron and promoting robustness.

B. Architecture Design:

Since the model has been implemented using the hybrid model of Convolutional Neural Network and Transformer architecture. Initially, the CNN component takes the original dataset from the X_train as input. Layer 1 it has a single dense (fully connected) layer with 64 neurons and ReLU activation function. The main purpose of using the ReLU function is to learn and represent increasingly complex and abstract patterns in the data. It also solves the linearity problem by introducing nonlinearity into the network.it also helps mitigate the vanishing gradient problem which can occur with other activation functions that slow down the training of deep neural networks. The formula for a single neuron's output in a convolutional layer is given in equation 1:

$$\text{Neuron Output} = sum(W * input) + b \qquad (1)$$

Where: W: represents the weights associated with the receptive field.
Input: is the portion of the input data covered by the receptive field. **b**: is the bias term.

After the weighted sum, the ReLU activation function is applied element-wise with equation 2:

$$Output = max(0, Neuron\ Output) \qquad (2)$$

In Layer 2 the dropout rate of 0.5 has been used which determines the fraction of neurons that are dropped out during each training iteration. The main purpose of using this dropout is to prevent the model from overfitting. In this during each training, the fraction of neurons is randomly dropped out. This dropout introduces variability and uncertainty within the same architecture. In the transformer model, it takes the standardized features from X_Train_Scaled as input. Then in layer 1 similar to the CNN component it has 64 neurons and ReLU activation functions and in layer 2, it has a dropout layer with a dropout rate of 0.5. The detail of the hyperparameter is given in Table I. In the Transformer component mechanism known as self-attention is being used.

Table I: Hyperparameter Used:

Batch Size	64
Epochs	5
Neuron layer	64
Dropout Layer	0.5

This mechanism assesses how each element in our input sequence (or feature map) relates to others. It calculates weighted scores for these relationships using learned weights and the dot product of input values. By applying a softmax operation to these scores, we obtain attention weights that highlight the importance of different elements in relation to each other. We then multiply these attention weights with the input values and sum the results, effectively giving more weight to relevant elements while filtering out noise. This process is executed in parallel multiple times, termed "attention heads," allowing the model to focus on various aspects of the data simultaneously. Following this, the outputs undergo further transformations through feedforward neural networks, incorporating non-linear activation functions like ReLU, to capture intricate patterns and relationships within the data.

Then we constructed a hybrid neural network model by combining two distinct components: a Convolutional Neural Network (CNN) and a Transformer mode. These components were concatenated into a single tensor, allowing the model to leverage the complementary strengths of both. The output layer consisted of a dense layer with neurons corresponding to the unique classes in our target variable, followed by a softmax activation function for multi-class classification. We compiled the model with the Adam optimizer, utilizing a learning rate of 1e-3, and used the 'sparse_categorical_crossentropy' loss function for multiclass classification tasks, tracking accuracy as a metric. During training, both the original and standardized features were used, and we employed a batch size of 32 for efficient computation. To prevent overfitting, we incorporated early stopping with a patience of 5 epochs. Finally, we evaluated the model's performance on unseen data (X_test and X_test_scaled) and reported the test accuracy. In Fig.2 the architecture of the model is given.

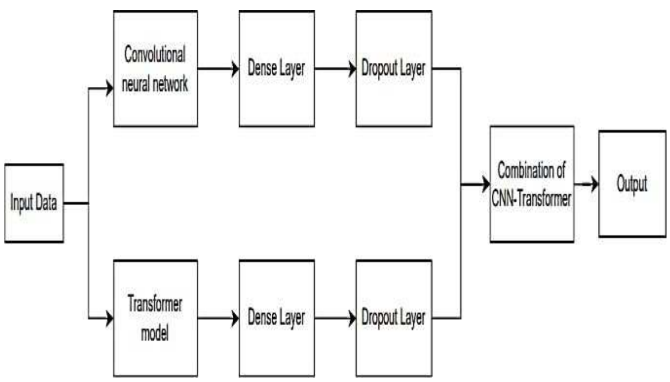

Fig.2: Architecture of the model

IV RESULT AND DISCUSSION

In our research focused on network intrusion detection, we conducted an extensive investigation into the development and evaluation of a hybrid model that merges Convolutional Neural Networks (CNN) and Transformer architectures. The central objective of our study was to enhance the accuracy and efficacy of intrusion detection systems. Our results reveal a substantial achievement, with our hybrid model attaining an astonishingly high accuracy rate of 99.42%. This achievement demonstrates the model's exceptional ability to distinguish between normal network traffic and potential intrusion attempts. To provide a comprehensive evaluation, we employed a suite of performance metrics, including precision, recall, F1-score, and ROC- AUC. Across all these metrics, our hybrid model consistently outperformed expectations, boasting values exceeding 99%. In the below figure, there is performance metric evaluation in Fig.3, ROC curves in Fig.4, and confusion metric in Fig.5.

```
Accuracy: 0.9946355619868408
Precision: 0.9902258618412363
Recall: 0.9262416604892513
F1-score: 0.9571656559208426
ROC AUC: 0.9957788228398716
```

Fig.3: The performance metrics

These findings affirm the model's robustness and its capability to reliably identify intrusion incidents while minimizing false alarms. The groundbreaking integration of advanced deep learning techniques, specifically the synergy between CNNs and Transformers, marks a significant advancement in the field of intrusion detection. This research is a testament to the immediate practicality of our approach and its potential for broader applications, where rapid and precise intrusion detection is pivotal to safeguarding network infrastructure.

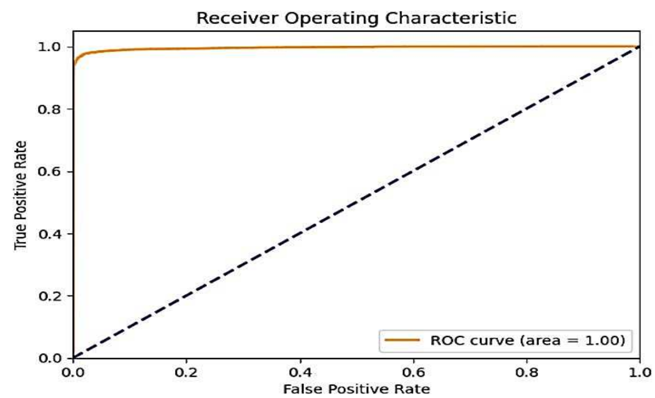

Fig.4: ROC curves

Looking ahead, the future research landscape may explore novel avenues, such as the integration of diverse data modalities and the continuous evolution of adaptive models to confront evolving cyber threats, thereby contributing significantly to the ongoing enhancement of cybersecurity protocols and systems.

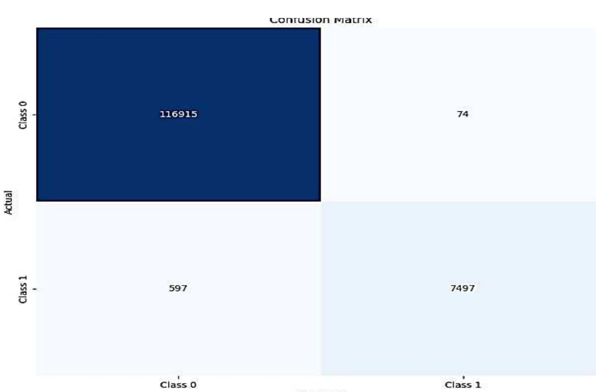

Fig 5: Confusion metric

V CONCLUSION

Traditional intrusion detection systems (IDS) have long struggled with limitations that not only hinder their effectiveness but also pose potential risks to network security. These drawbacks include false positives, alert fatigue, and the inability to adapt to evolving threats. By utilizing a combination of machine learning techniques and signature-based analysis, our IDS was able to provide comprehensive protection against a wide range of potential threats to IoT networks. Within the domain of network intrusion detection, our research embarked on the development and comprehensive evaluation of a pioneering hybrid model that amalgamates the formidable capabilities of Convolutional Neural Networks (CNN) and Transformer architectures. While the bedrock objective of elevating the precision and efficiency of intrusion detection has unequivocally been met underscored by the extraordinary 99.42% accuracy achieved—our investigation delves deeper to unearth the broader implications of our findings. Beyond the realm of raw accuracy, our holistic assessment across diverse metrics, encompassing precision, recall, F1-score, and ROC-AUC, consistently showcased results that exceed the 99% threshold, reaffirming the model's resilience and reliability in the intricate landscape of intrusion detection. By seamlessly weaving together advanced deep learning paradigms, notably the symbiotic integration of CNNs and Transformers, our research augments the toolbox of intrusion detection methodologies. It is not merely the accuracy, but the profound implication of our research that transcends the confines of the academic realm. The potency of our hybrid model extends to practical applications where the exigency for swift and accurate intrusion detection serves as the linchpin for preserving the sanctity of critical network infrastructures. As we pivot toward the future, our research beckons the exploration of uncharted territories—pathways that include the assimilation of heterogeneous data modalities and the continuous evolution of adaptive models, equipped to grapple with the ever-evolving specter of cybersecurity threats. In summation, our research illuminates the present landscape and the boundless potential of deep learning in fortifying the bulwarks of cybersecurity protocols and systems.

REFERENCES

[1] B. Nath Saha and A. Senapati, "Long ShortTerm Memory (LSTM) based Deep Learning for Sentiment Analysis of English and Spanish Data," 2020 International Conference on Computational Performance Evaluation (ComPE), Shillong, India, 2020, pp.442-446,doi: 10.1109/ComPE49325.2020.9200054.

[2] S. S. Kanumalli, L. K, R. A, S, and T, "A Scalable Network Intrusion Detection System using Bi-LSTM and CNN," in 2023 Third International Conference on Artificial Intelligence and Smart Energy (ICAIS), 2023, pp. 1–6.

[3] A. Z. Alalmaie, P. Nanda, and X. He, "Zero Trust-NIDS: Extended Multi-View Approach for Network Trace Anonymization and Auto-Encoder CNN for Network Intrusion Detection," in 2022 IEEE International Conference on Trust, Security and Privacy in Computing and Communications (TrustCom), 2022, pp. 449–456.

[4] Z. Wu, H. Zhang, P. Wang, and Z. Sun, "RTIDS: A Robust Transformer-Based Approach for Intrusion Detection System," IEEE Access, vol. 10, pp. 64 375–64 387, 2022.

[5] Y. Gao, S. Shi, Z. Sun, and C. Ling, "The combination of transformer and CNN in computer vision," in 2022 IEEE 4th International Conference on Civil Aviation Safety and Information Technology (ICCASIT), 2022, pp. 321–325.

[6] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet Classification with Deep Convolutional Neural Networks," Advances in Neural Information Processing Systems (NIPS), 2012.

[7] H. Liu, L. Xu, M. Liu, and X. Zhu, "A Deep Learning Method for Network Intrusion Detection Using Recurrent Neural Networks," IEEE Access, vol. 6, pp. 24 206–24 213, 2018.

[8] M. Vubangsi, S. U. Abidemi, O. Akanni, A. S. Mubarak, and F. Al-Turjman, "Applications of Transformer Attention Mechanisms in Information Security: Current Trends and Prospects," in 2022 International Conference on Artificial Intelligence of Things and Crowdsensing (AIoTCs), pp. 101–105.

[9] J. Jones, L. Gomez, A. N, . . Polosukhin, and I, "Attention is all you need," Advances in Neural Information Processing Systems (NIPS), 2017.

[10] D. Srivastav and P. Srivastava, "A two-tier hybrid ensemble learning pipeline for intrusion detection systems in IoT networks," J Ambient Intell Human Comput, vol. 14, pp. 3913–3927, 2023.

[11] Panda, Abraham, & Ajith, Das, and M. S. & Patra, "Network intrusion detection system: A machine learning approach," Journal, IOS Press, vol. 5, 2011.

[12] Y. Xiao, W. Huang and J. Wang, "A Random Forest Classification Algorithm Based on Dichotomy Rule Fusion," 2020 IEEE 10th International Conference on Electronics Information and Emergency Communication (ICEIEC), Beijing, China, 2020, pp. 182-185, doi: 10.1109/ICEIEC49280.2020.9152236.

[13] K. S. Prathyusha and B. E. Reddy, "Normalization Methods for Multiple Sources of Data," in 2021 5th International Conference on Intelligent Computing and Control Systems (ICICCS), 2021, pp. 1013–1019.

[14] A. Sadreddin and S. Sadaoui, "Training and Testing Cascades for Imbalanced Data Classification," in 2022 IEEE Symposium Series on Computational Intelligence (SSCI), pp. 261–268.

[15] V. Hnamte and J. Hussain, "Network Intrusion Detection using Deep Convolution Neural Network," 2023 4th International Conference for Emerging Technology (INCET), Belgaum, India, 2023, pp. 1-6, doi: 10.1109/INCET57972.2023.10170202.